

A smile can go a long way: The effects of dominant and rewarding smiles of coaches on athletes in an evaluative performance context

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Abstract

Non-verbal expressions from other people play an important role in everyday life. We tested the psychological and physiological effects of different non-verbal facial expressions given by a coach to athletes in a performance context. In a study with 60 athletes, we tested how dominance versus reward smiles shown by a coach after various performances affected athletes' subjective affective state, stress levels, and their own facial expressions. Results showed that athletes who perceived dominant smiles after their performances had higher heart-rates than those who received rewarding smiles. Athletes receiving dominant smiles also reported feeling less happy after their performances than athletes receiving rewarding smiles. No effects were evident on subjective stress levels and facial expressions of emotion measured by Noldus FaceReader. The present results are the first to show that subtle differences in facial expressions of coaches can affect an athlete's heart rate and affective state in an evaluative performance context. The results join research on the interpersonal effects of non-verbal behaviour between athletes and their coaches. They provide evidence that the facial expressions of coaches have communicative content that is perceived by the athletes and influences them on different levels.

Keywords

Affective state, body language, emotion, facial expression, non-verbal behaviour, stress

Introduction

Although most people would agree that athletes perceiving emotional expressions in their coaches would be affected by them, surprisingly, little research has investigated the effects of coaches' emotional expressions on athletes. Therefore, the present research was grounded within the facial expression literature and attempted to transfer recent psychological findings to the context of sport, which has shown that even subtle variations of facial expressions can have a profound impact on perceivers. More specifically, we tested the psychological and physiological effects of different forms of non-verbal facial expressions displayed by a coach to athletes in a performance context.

Non-verbal feedback from other people plays an important role in everyday life, presumably because humans are confronted with this from very early developmental stages on.¹ The interaction between parents and infants relies heavily upon non-verbal feedback in the form of exchanging non-verbal signals. Also, at later developmental stages, humans continue to rely on the non-verbal feedback they get from other people in a variety of situations, such as

educational settings. In this respect, facial expressions² have been shown to be of particular importance as these have evolved to be a powerful signalling system that facilitates the communication of vital social information. Within facial expressions, the human smile has fascinated people particularly and is known to affect numerous social outcomes.³ However, this form of feedback has rarely been studied amongst adults. Hence, the present research attempted to address this shortcoming by investigating the

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effects of different forms of the human smile on athletes in an evaluative performance context, which can arguably be considered a frequently occurring form of feedback in coaching situations in sports.

The popular saying ‘a smile goes a long way’ summarises a growing body of research showing that the human smile is a powerful ‘social tool’ in influencing human interaction.⁴ While the contraction of a single muscle, the zygomaticus major, is considered to be sufficient to qualify a facial expression as a smile, research has shown that not every smile is created equal and that the smile varies across various different dimensions, probably more so than any other human facial expression (e.g. Refs.^{5,6}). The most prominent type of smile has been labelled Duchenne smile⁷ characterised by the appearance of crow’s feet around the eyes. This label was introduced to distinguish authentic smiles, which reflect true positive affect, from fake smiles, which are shown in social situations but do not reflect true positive affect.⁸ However, this true–false classification of smiles has proven problematic on various accounts. Therefore, the present research builds on more recent theorising⁴ that is derived from behavioural ecology.⁹ This theoretical position has been specified in the simulation of smiles (SIMS) model,¹⁰ which proposes that distinct human smiles exist in the service of meeting three basic tasks in the social lives of humans, that is, to reinforce desired behaviours in oneself and in others (the reward smile); to facilitate the formation and maintaining of relationships (the affiliation smile); and to establish and negotiate social hierarchies (the dominance smile).

In the present research, we focused on the effects of reward and dominance smiles, as these seem to be of particular importance in the context of an athlete receiving

feedback from a coach in an evaluative performance situation. Figure 1 shows an example of a stereotypical dominant smile on the left and a reward smile on the right. Recent research (see Ref.⁴ for a review) indicated that eyebrow flashes – involving the Inner and Outer Brow Raiser – and a symmetrical contraction of the zygomaticus major are perceived as rewarding by participants. This type of facial configuration has been shown to have the potential to reinforce a person’s behaviour by activating reward centres in a perceiver’s brain.¹¹ Stated differently, reward smiles can induce pleasurable states in the perceivers and thereby reinforce the behaviour that gave rise to the smile initially. On the other hand, unilateral, asymmetrical zygomaticus major movements and action units that have been related to disgust displays including the Nose Wrinkler and Upper Lip Raiser are perceived as dominant. This type of facial configuration has been suggested to communicate an intention to destabilise, or maintain asymmetric hierarchical relations between the displayer and the observer. It has further been suggested that dominance smiles came about in humans as a tool for managing hierarchical relationships (e.g. challenging higher status or subduing lower status) without a substantial risk of physical violence.⁴

Pertinent to the present study, there is reason to believe that non-verbal feedback in the form of the two aforementioned smile types has the potential to impact perceivers in evaluative performance situations, as non-verbal feedback has been shown to be experienced as spontaneous and honest reflections of internal evaluations.¹ The limited research^{12–14} that has investigated the non-verbal communication of evaluative feedback has shown that people who received positive non-verbal signs of evaluation (e.g. smiles) showed lower physiological activity than people



Figure 1. Sample frames of the video stimuli used in the present research. On the left an example of a dominant smile and on the right an example of the reward smile.

who received negative non-verbal signs of evaluation (e.g. frowns). Of particular importance to the present research the only study we are aware of that has investigated the impact of different smile types in an evaluative performance context was conducted by Martin and colleagues.¹⁵ The study has demonstrated that various smiles can augment or dampen hypothalamic–pituitary–adrenal (HPA) axis activity in an evaluative context.¹⁵ The HPA has been shown to be a major player within the neuroendocrine system that controls bodily reactions to stress. More specifically, this research has shown that dominant smiles increased HPA axis activity (increased heart rate and salivary cortisol) and reward smiles seemed to buffer against higher physiological activity (decreased heart rate and salivary cortisol) in an adapted version of the Trier Social Stress Test.¹⁶

While initial research in sports has begun to investigate how an athlete's behaviour and performance can potentially be affected by a coach's emotional expressions,^{17–22} we are not aware of any experimental research that has investigated the effects of coaches' facial expressions on an athlete's physiology and current affective state when trying to perform at their best. Previous research in the field of sport coaching has mainly focused on verbal feedback as this represents one of the main behaviours occurring in sport coaching.²³ Within this literature verbal feedback has been described as a frequently used and high-impact strategy to enhance performance of athletes.^{24–27} However, verbal feedback is obviously not the only coaching behaviour occurring during training and competition that has an impact on athletes. Pertinent to the present research, an important study²² on the interpersonal effects of emotions has provided first evidence on the effects of coaches' emotional expressions on athletes during sport competitions. Specifically, this study demonstrated that coaches' emotional expressions of happiness and anger were associated with players' experiences of happiness and anger, and consequently affected performance of athletes. These findings are in line with additional empirical research showing that coaches' non-verbal behaviour affects athletes' confidence^{28–30} and other affective states.^{20,31}

The present research attempted to build and extend previous research by investigating the impact of evaluative non-verbal feedback of coaches in the form of reward and dominant smiles during stressful cognitive (in the form of a speech) and physical tasks amongst athletes. As athletes are constantly performing in the presence of coaches with their performance being evaluated, we considered this context well-suited to gain further insights into the potential impact of different smiles displayed by a coach. Given that coaches and athletes are connected in an authority figure–subordinate relationship,²⁰ we consider the reward and dominance smile particularly interesting as coaches can sometimes be motivated to reward athletes' behaviour or maintain asymmetric hierarchical relations with athletes

via non-verbal feedback. For example, a coach might show rewarding smiles after an athlete has shown a good performance. On the other hand, a coach might show a dominant smile if an athlete has shown disrespect or does not show effort during training. As previous research has shown that people are very receptive to subtle differences in facial expressions^{1,4,9–11} and the effects of coaches' specific facial expressions have not yet been systematically examined in an experimental context we tested if athletes would be differentially affected by dominant and reward smiles.

To provide a general model on the effects that emotional expressions like the aforementioned smile types can have on observers, van Kleef³² proposed the emotions as social information model (EASI-model) to specify how non-verbal expressions may exert interpersonal effects. Central to the model is the assumption that non-verbal expressions affect observers via two routes: Inferential processes and/or affective reactions. Inferential processes describe how an observer of the expression is able to infer certain information about internal states (e.g. feelings, attitudes, motivation, cognitions) of other people and the current situation. Applied to the present research, athletes observing the coach's smile might make the inference that the coach is happy about their performance, which in turn might affect both their physiology and feelings. In addition, the affective reactions route might arguably also be relevant for the present research as previous research (summarised in Ref.³²) has indicated that emotional expressions might have a more direct affective effect on observers, for example via emotional contagion. Hence, it seems reasonable that athletes' physiology and their own facial expressions might be affected when observing different smiles from a coach.

To test this line of reasoning, we attempted to create a realistic scenario in which athletes had to convince a coach in a speech about their physical and mental strength and perform in a difficult physical task while being evaluated by a coach. One group of participants received, at various instances, reward smiles from the coach, whereas the other group received dominance smiles. The between-subject design was based on previous research¹⁵ to avoid demand effects that might have occurred in a within-subject approach. As Martin and colleagues showed that dominant smiles increased HPA axis activity compared to reward smiles, we expected that athletes perceiving dominant smiles as compared to reward smiles after their performances would have a higher heart-rate and report feeling more stressed, and experiencing more positive affect. This hypothesis was based on the EASI theorising³² and recent empirical evidence suggesting that dominant and reward smiles, delivered as evaluative feedback, influence perceivers' HPA axis activity in a manner congruent with their distinct social meaning.¹⁵

We assumed that performing in front of a coach would be a stressful situation for athletes. We further predicted

that receiving rewarding facial feedback (reward smile) in this situation would lead to a decrease in HPA activity and, in turn, to a lower heart-rate and less-subjective feelings of stress and more positive affect. Whereas the perception of a dominant smile by a coach in this stressful situation should amplify HPA activity and as a consequence increase heart-rate and subjective feelings of stress and negative affect. More specifically, these hypotheses were based on the inferential processing route specified by the EASI model, because athletes would likely infer that they had been performing well/poorly when observing either the rewarding or dominating smile of the coach after their respective performances and this would likely differentially affect their own emotional states.

In addition, we explored the facial non-verbal behaviour of athletes via Noldus FaceReader as a reaction to receiving different kinds of non-verbal feedback from coaches, because it seems feasible that higher stress levels in response to perceiving dominant smiles in comparison to reward smiles would also show on an athlete's face. This reasoning was based on a large body of evidence showing that non-verbal facial expressions often correspond with internal states (e.g. Refs.^{6,9}). Further, evidence on emotional contagion as specified within the affective processing route in the EASI model³² might also suggest that athletes' facial expressions might change as a consequence of observing different facial expressions by their coach. However, we consider this less likely given the authority figure-subordinate relationship between coaches and athletes,³² which makes the inferential processing mechanism more likely than the affective processing mechanism.³²

Method

We tested whether dominant and reward smiles from the coach influenced physiological stress responses (change in heart rate), subjective feelings of stress and affect, and facial expressions of the observing athletes that were consistent with their different social meanings. A MANOVA, a series of follow-up ANOVAs, and *t*-tests were calculated for this purpose.

Participants

Sixty participants (33 male, 27 female; $M_{\text{age}} = 22.35$ years; $SD = 4.22$ years) participated in the study. All participants were recreational or semi-professional athletes (16 track and field; 7 endurance running; 13 tennis; 11 team ball sports like soccer or volleyball; the remaining 13 came from less mainstream sports like fencing, gymnastics, judo, swimming, etc.). On average, the athletes reported to spend 6 h/week ($SD = 2.93$) on their respective sports. Participants gave written informed consent before the experiment. 29 (14 female; $M_{\text{age}} = 23.72$ years; $SD = 4.70$ years) were randomly assigned to the 'reward-smile-group'

and 31 (13 female; $M_{\text{age}} = 21.06$ years; $SD = 3.30$ years) to the 'dominant smile group'. Pre-inclusion criteria limited participation to male and female participants without a diagnosed heart condition and not currently taking medications that alter the cardio-vascular function or hormone levels. Participants were further instructed to refrain from alcohol and caffeine consumption within 24 hours of their participation. Sample sizes were based on previous comparable research,¹⁵ and were powered to detect large-to-medium effects ($f = 0.24$ in a sensitivity analysis with a sample size of 60 and power of .95 in a 2×2 ANOVA for the within-between subject interaction) in the calculated ANOVAs. The study protocol was approved by the local university's ethics committee.

Procedure and measures

The experimental protocol was adapted from Martin et al.¹⁵ After participants had provided informed consent, they completed a questionnaire assessing their compliance with pre-restriction criteria (e.g. alcohol and caffeine use) as well as other personal and medical information (e.g. prescription drug use). Then the experimenter attached sensors to the participant's chest to measure aspects of their cardio-vascular reactivity throughout the experiment. In addition, participants were video-recorded to act as an additional stressor³³ throughout the experimental procedure, both for analysis purposes with the Noldus FaceReader software.

First, a 5-minute baseline measure of the participant's heart activity was recorded. This measure was taken while participants were standing, as they would also be standing during the experimental protocol. During this 5-minute period, participants also rated their affect and subjective stress level on two visual-analogue scales. Then participants were informed to step into their athlete roles and were asked to prepare and give a 4-minute speech about their physical and mental strengths and to perform in two demanding physical tasks. In addition, they were told that all their performance would be monitored and evaluated by a high-level coach via Skype as the coach could not be physically present due to COVID-19 restrictions at the time. The experimenter however was allowed to be present during the entire testing period. They were further told that the goal of the athletes was to convince the coach that they should make the cut for the national or state team. Analogous to the approach of Martin et al.¹⁵ the experimenter then introduced a facial expression recognition software (Noldus FaceReader) and the participants were informed that the program extracted meaningful facial expressions from the live video feed showing the coach. To increase credibility of the Skype live stream and the experimental manipulation the coach welcomed the participants in a Skype live stream (similar to the procedure of Martin et al.¹⁵) and informed the participants about the following procedure. He informed them that he

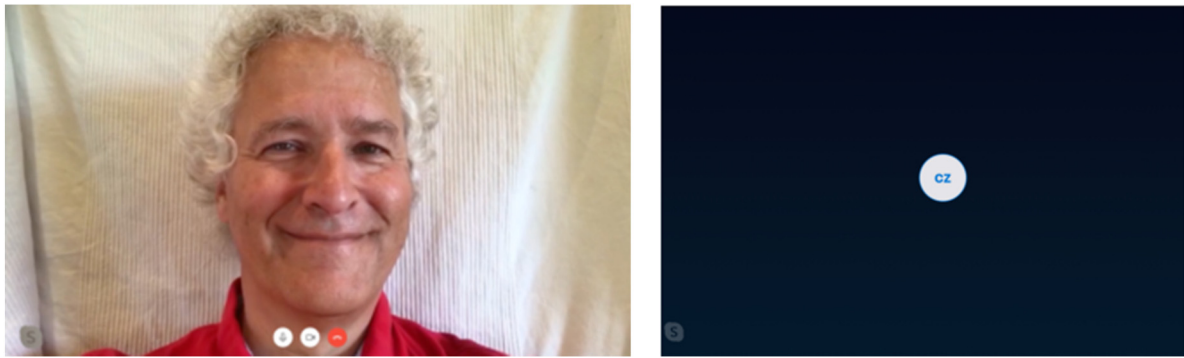


Figure 2. Sample frames of the Skype video stream. On the left an example of a non-verbal feedback video and on the right an example of the skype stream during the athletes' performances.

would be present during the entire time and be evaluating the athlete's performance online but would turn off his webcam and sound during the performance to avoid distraction. Figure 2 illustrates the Skype video stream which the athletes saw: On the left during a pre-recorded feedback video and on the right, the black Skype screen with webcam and sound turned off during the performances.

The first task of the athletes was to perform the 4-minute speech to convince the coach about their personal strengths. Then the Skype live stream started, and the coach explained to the athletes that he would evaluate their task performances. He also told them that he would evaluate how assertive and confident their facial expressions and body language appeared, and how well they convinced him as a coach that they are the proper athlete for the team. Moreover, he added that he will talk to them again after the tasks.

Then the coach was asked by the experimenter to turn off his webcam and sound. Subsequently, the participants commenced their speech. After every minute of the speech or after the end of a sequence, when athletes stopped talking about one of their strengths, the participants were asked to look at the screen to observe the emotional feedback of the coach. This feedback was, in fact, the pre-recorded smiles of the coach that appeared to be live on the Skype screen (which participants were told would be analysed by the Noldus FaceReader software for emotional content). Similar to Martin et al.,¹⁵ they were further told that the expressions displayed by the coach showed the reactions of the coach while watching their speech and physical tasks. Hence, 4 smile videos were presented after every minute of the speech to the participant (4 reward smiles in the 'reward group' and 4 dominant smiles in the 'dominant group'). After the speech, participants again filled out the two visual analogue scales.

Next, participants performed the physical tasks (see Figure 3). After the completion of both physical tasks, participants filled out the two visual analogue scales assessing their subjective stress level.

Performance tasks. Speech. The first task of the athletes was to perform the 4-minute speech in which they should speak in 1-minute blocks about athletic and mental strengths (1 minute per individual strength) and try their best to convince the coach that they were the right candidate for a state or national team (depending on the profile of the respective athlete). Participants were also informed that the coach would evaluate how convincing and assertive their speech was. Participants had 5 minutes to prepare their speech. They were not given a pen or paper in preparation for their speech but were asked to prepare the speech in their mind.

Physical task. In this task, participants were holding two tennis balls in both hands and had to throw them up simultaneously, clap their hands while the balls were in the air, and then catch them again while crossing the hands without the balls falling on the ground. After that, the balls had to be thrown up simultaneously out of the crossed-hands position, clap hands again, and then catch them with the hands switched back to the parallel starting position. This task was pre-tested to be a difficult physical task that induced a moderate level of stress in participants. After an explanation of the task, participants directly started with the task without practising. They were instructed by the experimenter that their goal was to complete the task with as few mistakes as possible (i.e. dropping the tennis balls). Participants had two trials lasting 30 seconds each. After each trial, participants were shown the non-verbal feedback of the coach again via the simulated Skype stream (i.e. two videos; either showing two dominant or two reward smiles).

The last physical task was a precision throwing task. Participants were asked to take 10 throws with a tennis ball and hit a bin (30-cm diameter, 70-cm high) from a distance of 4 m as often as possible. Participants were instructed to perform underarm (instead of overhead) throws. To increase the difficulty, participants were asked not to look at the target while trying to hit it. Again, participants performed two trials and received the feedback



Figure 3. First physical task. Top left column: starting position; Top right column: throw both balls in the air; Bottom left column: clap while balls are in the air; Bottom right column: catch both balls with the hands crossed.

videos after every trial. After the completion of the first trial participants were told that their aim for the second trial was to hit the target once more than in the first trial.

Experimental video manipulation. A trained actor played the role of the coach during the live Skype video stream and for the video material of the smile feedback. Initially, the male actor (58 years old, see Figure 1) was recorded in 60 videos showing 30 dominance and 30 reward smiles based on the coding descriptions in the literature.³⁴ No other head movement or non-verbal behaviour besides the two smile types was visible in the video material. All videos were shown without sound. In a pilot study via the online platform SoSci Survey 20 participants rated the videos on continuous scales (0–100) in terms of dominance and reward. Based on the analyses of these ratings, 16 videos ($M_{\text{duration}} = 4.19$ seconds; $SD = 0.56$ seconds; 8 per smile category) were selected that were rated most clearly in terms of dominance and reward (highest ratings on dominance scale for the 8 dominant smile videos and highest

ratings on the reward scale for the 8 reward videos; the videos differed significantly in terms of dominance $p = .001$ and reward $p = .001$ ratings).

Measures. Demographic questionnaire. Several initial items assessed the age, gender, sport-specific questions (what type of sport, level, hours of training per week), and the athlete's general level of performance anxiety (7-point Likert scale: 1 = not at all anxious; 7 = highly anxious).

Heart rate. Heart rate was monitored with the device Garmin Fenix 5× Plus. The mean of the 5-minute period at the start of the procedure was taken as a baseline measure. Further, we calculated differential heart rate values after receiving the non-verbal feedback to test the hypothesis. To this end, we subtracted the mean of a 10-second period directly after receiving the non-verbal feedback from the baseline measure. This procedure is derived from evidence showing that perceived stress

hardly delays an individual's heart rate.^{33,35} Given that every athlete received non-verbal feedback from the coach at eight different times, we calculated eight different heart rate measures in addition to the baseline measure. Finally, we also computed one omnibus heart rate reactivity measure by taking the average of all 8 feedback measures and subtracting it from the baseline measure.

Visual analogue scales for stress and affect. Two 7-point Likert scales were administered at three measurement times (during the baseline heart rate monitoring, after the speech, and after the two physical tasks). The first scale assessed the subjective affect illustrated with smiley emoticons, where 1 indicated very negative affect (both corners of the mouth were pulled down) and 7 an very positive affect (both corners of the mouth were pulled up). The second scale assessed their subjective stress level rating from 1 (not at all) to 7 (very). Due to experimenter error data went missing for three participants for the visual analogue scales.

Action unit (AU) intensities. Participants were filmed with an HD video camera from a frontal perspective during the entire protocol. From this video stream, 9 different videos were cut analogous to the heart rate data, resulting in nine different video clips that were analysed with Noldus FaceReader: one 5-minute baseline video and eight 10-second videos after receiving the non-verbal feedback from the coach. Due to poor video quality, too much participant movement, or experimenter error, not all video clips could be analysed by FaceReader. Altogether 485 clips were suitable for the FaceReader analysis.

Using Active Appearance Modelling,³⁶ FaceReader calculates the activation intensity of 20 distinct muscle groups in the face (i.e. 'action units'), which are used to generate facial expressions.² The validity of coding action unit intensities has been reported as acceptable for most action units.³⁶ FaceReader automatically computes continuous variables reflecting the intensity of the following AUs shown in Table 1 (for description and illustration of individual AUs, see: <https://www.noldus.com/applications/facial-action-coding-system>):

Action Units were exported as continuous variables with FaceReader ranging from 0 to 1. Values from 0.00 to 0.16 are classified as Trace, .16–.26 as Slight, .26–.58 as Pronounced, .58–.90 as Severe, and .90–1 as Max Intensity for every video frame.^{2,36} The dependent variable was the mean of every analysed frame in the respective video.³⁷

Emotional facial expressions. Subsequently, whole-face expressions were classified into seven basic emotions (Happiness, Sadness, Anger, Surprise, Fear, Disgust, and Neutral) by the FaceReader algorithm, following Ekman and Friesen's Facial Action Coding System (for the validity of this classification procedure, see Refs.^{37,38}). We used the default settings using a general face model with continuous calibration. The dependent variable was the mean of the respective basic emotion of every analysed frame in the respective video.

Valence and arousal. In addition, a combination of action unit intensity and intensity of positive vs. negative emotion is used to infer general levels of valence and arousal (i.e. based on Russell's circumplex model of affect³⁹). The dependent variable was the mean of the valence and arousal scores in every analyzed frame in the respective video.

Table 1. Specific action units and their description.

Action unit	Description
AU 1	Inner Brow Raiser (Muscular basis: frontalis, pars medialis)
AU 2	Outer Brow Raiser (Frontalis, pars lateralis)
AU 4	Brow Lowerer (depressor glabellae, depressor supercilii, and corrugator supercilii)
AU 5	Upper Lid Raiser (levator palpebrae superioris, and superior tarsal muscle)
AU 6	Cheek Raiser (Orbicularis oculi, pars orbitalis)
AU 7	Lid Tightener (Orbicularis oculi, pars palpebralis)
AU 9	Nose Wrinkler (Levator labii superioris alaeque nasi)
AU 10	Upper Lip Raiser (Levator labii superioris, caput infraorbitalis)
AU 14	Dimpler (Buccinator is the underlying muscle)
AU 15	Lip Corner Depressor (Depressor anguli oris)
AU 17	Chin Raiser (mentalis)
AU 18	Lip Pucker (incisivii labii superioris and incisivi labii inferioris)
AU 20	Lip Stretcher (risorius w/ platysma)
AU 23	Lip Tightener (orbicularis oris)
AU 24	Lip Pressor (orbicularis oris)
AU 25	Lips Part (depressor labii inferioris, or relaxation of mentalis or orbicularis oris)
AU 26	Jaw drop (masseter; relaxed temporalis and internal pterygoid)
AU 27	Mouth Stretch (pterygoids and digastric)
AU 43	Eyes Closed (relaxation of Levator palpebrae superioris)

Transparency and openness

We report how we determined our sample size, all data exclusions (if any), all manipulations, and all measures in the study, and we follow JARS. All data, analysis code, research materials, and all data and materials used in this study will be made available at reasonable request.

Results

Multivariate analysis on heart rate, affect, and stress

As a first omnibus hypothesis test, we ran a 2 (baseline vs. after-feedback) $\times$ 2 (dominant vs. reward smile) mixed-design MANOVA on the heart rate data, affect, and subjective stress data (It was not possible to include the FaceReader data in this MANOVA as this would have been too many variables). Therefore, we initially averaged the after-feedback data to obtain one value for every participant as a proxy for their after-feedback heart rate (8 measurement times, see Figure 4), after-feedback affect (2 measurement times, see Figure 5), and their after-feedback stress (2 measurement times, see Figure 6). This 2 (baseline vs. after-feedback) $\times$ 2 (dominant vs. reward smile) mixed-design MANOVA using Pillai's trace revealed a significant main effect of measurement time ($V=.302$; $F(3, 53)=7.646$, $p<.001$, $\eta_p^2=.302$). The main effect for group

was not significant ($V=.037$; $F(3, 53)=0.672$, $p=.573$, $\eta_p^2=.037$). Most importantly, there was a significant interaction between measurement time and group ($V=.235$; $F(3, 53)=5.431$, $p=.002$, $\eta_p^2=.235$). To further illuminate the significant multivariate effects, we ran further univariate analysis on the individual dependent variables.

Heart rate data

The descriptive results are displayed in Figure 4 (for the individual feedback measures). A 2 (baseline vs. after-feedback) $\times$ 2 (dominant vs. reward smile) mixed-design ANOVA on the mean baseline and average feedback heart-rate measures revealed a significant main effect of measurement time $F(1, 57)=16.44$, $p<.001$, $\eta_p^2=.023$. The main effect for group was not significant, $F(1, 57)=1.64$, $p=.21$, $\eta_p^2=.028$. Most importantly, there was a significant interaction between measurement time and group $F(1, 57)=10.75$, $p=.002$, $\eta_p^2=.159$. This significant interaction was supported by an independent t-test on the difference heart rate values between baseline and average feedback measure, $t(57)=3.28$, $p=.002$, two-tailed, $d=.854$ as the mean difference values were higher for the dominant group $M=-10.70$; $SD=11.63$ compared to the reward group ($M=-1.13$; $SD=10.74$). The gender of participants did not significantly moderate the pattern of results when entered as an additional between-subject factor or covariate

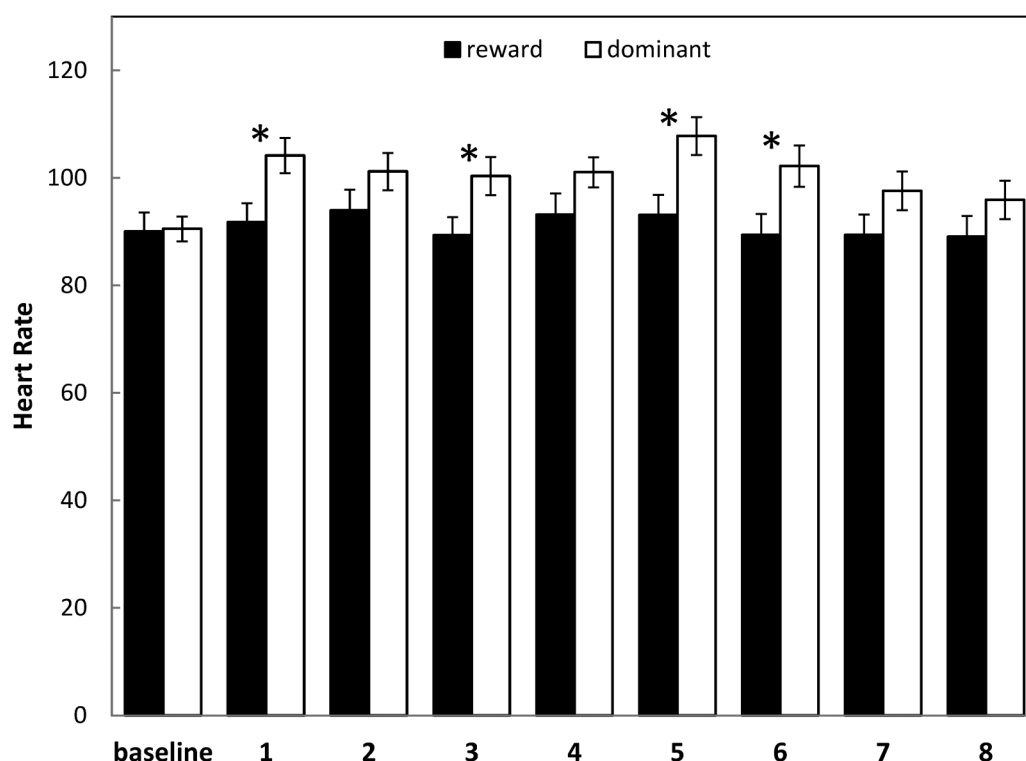


Figure 4. Heart-rate data as function feedback group and measurement time. Significant differences are marked with *. Error bars represent standard errors.

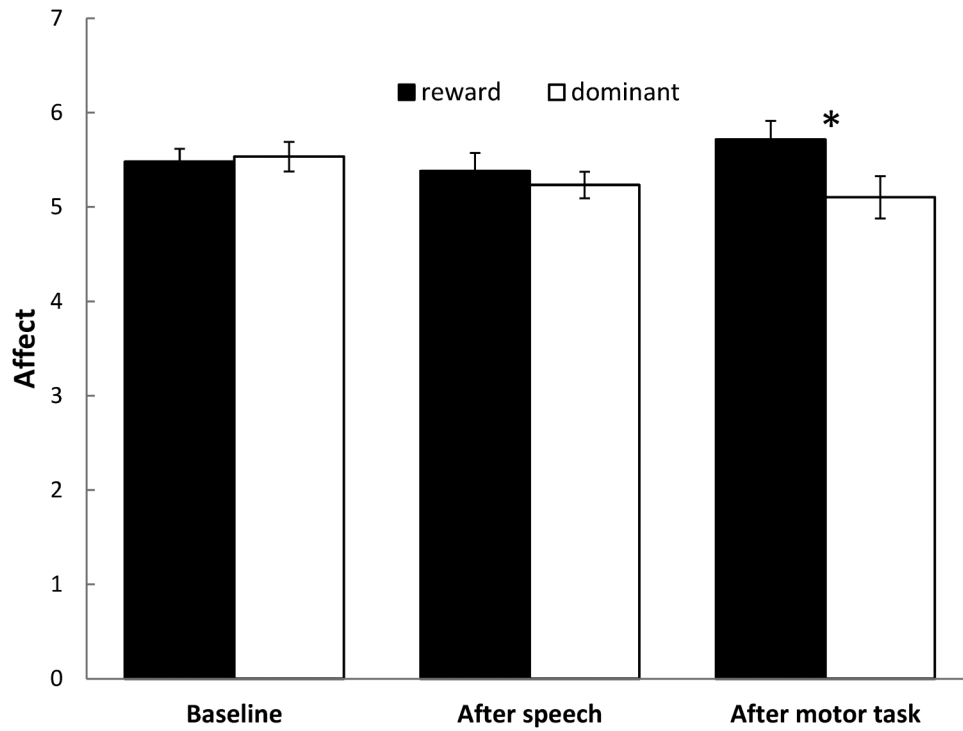


Figure 5. Affect data as function feedback group and measurement time. Significant differences are marked with *. Error bars represent standard errors.

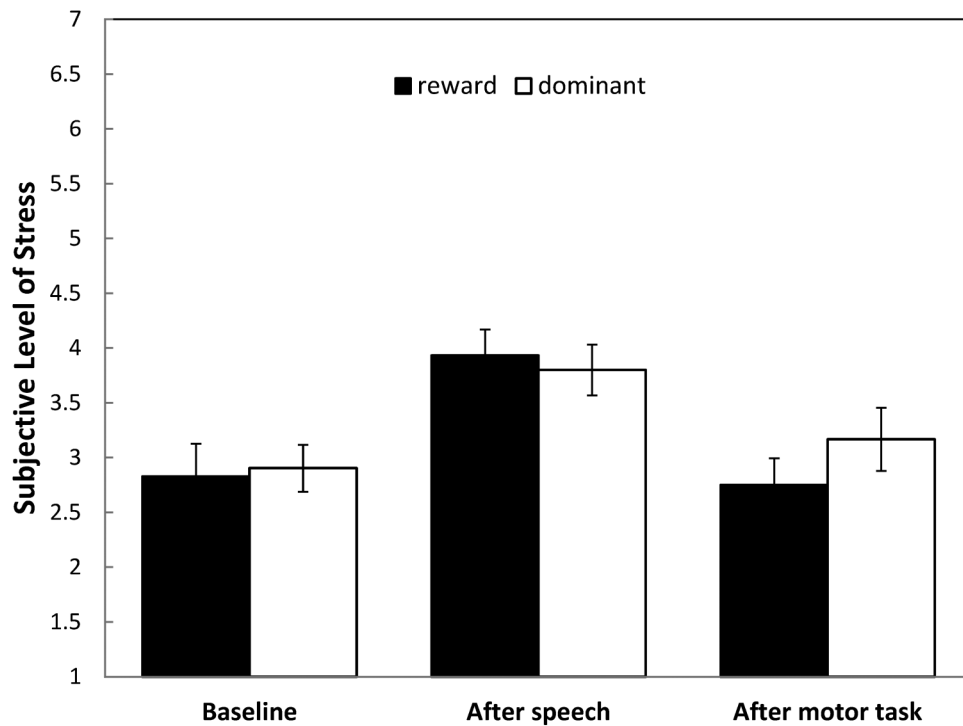


Figure 6. Subjective stress data as function feedback group and measurement time. Significant differences are marked with *. Error bars represent standard errors.

and there was no significant differences between males and females in difference heart rate values between baseline and average feedback measure, $t(57)=0.271$, $p=.788$, two-tailed, $d=.071$.

A further 2 (dominant vs. reward smile) $\times$ 9 (measurement time) ANOVA (see Figure 4) revealed a significant main effect of measurement time $F(8, 424)=4.836$, $p<.001$, $\eta_p^2=.084$ and for group, $F(1, 53)=5.094$, $p=.028$, $\eta_p^2=.088$. Again, there was a significant interaction between measurement time and group $F(8, 424)=2.265$, $p=.032$ (Huynh-Feldt correction), $\eta_p^2=.041$. Planned-contrast analyses showed significant differences at measurement point 1 ($p=.007$), 3 ($p=.016$), 5 ($p=.003$), 6 ($p=.012$) between the dominant and reward smile group, whereas measurement points 2 ($p=.087$), 4 ($p=.060$), 7 ($p=.063$), 8 ($p=.101$) did not reach significance.

Subjective affect and stress level

The descriptive results for affect are displayed in Figure 5. A 2 (dominant vs. reward smile) $\times$ 3 (baseline vs. feedback after speech vs. feedback after physical tasks) mixed-design ANOVA on the subjective affect ratings did not reveal a significant main effect for measurement time $F(1,76, 96,82)=1.11$, $p=.330$, $\eta_p^2=.020$ or group, $F(1, 55)=1.44$, $p=.235$, $\eta_p^2=.026$. There was, however a significant interaction between measurement time and group $F(1,76, 96,82)=3.73$, $p=.03$, $\eta_p^2=.063$, as affect levels decreased from baseline measurement and the two feedback measurements for the dominant feedback group and went slightly up in the reward feedback group. Planned-contrast analyses showed no significant differences at baseline ($p=.405$) or after the speech ($p=.272$) between the dominant and reward smile group, but did show a significant difference after the physical task ($p=.023$).

The descriptive results for subjective stress levels are displayed in Figure 6. A 2 (dominant vs. reward smile) $\times$ 3 (baseline vs. feedback after speech vs. feedback after physical tasks) mixed-design ANOVA on the subjective stress ratings did reveal a significant main effect for measurement time, $F(2, 112)=13.74$, $p<.001$, $\eta_p^2=.197$. The main effect for group $F(1, 56)=0.23$, $p=.634$, $\eta_p^2=.004$ and the interaction between group $\times$ measurement time $F(2, 112)=0.89$, $p=.44$, $\eta_p^2=.014$. were not significant. Hence, the non-verbal feedback did not differentially affect subjective stress ratings in participants.

Facial expressions of athletes

Table 2 shows the results of a 2 (baseline vs. average of feedback) $\times$ 2 (dominant vs. reward) ANOVA on the averaged FaceReader variables and the corresponding descriptive statistics. None of the micro and macro facial behaviours significantly differed as a function of group and measurement time (except for the action unit chin raiser). Therefore, the

pattern of results shows that participants' facial behaviour did not systematically change depending on what type of non-verbal feedback they received from the coach (at least as measured by the FaceReader algorithm).

Individual difference analyses

To examine if the gender, age, or performance anxiety moderated the pattern of results on any of the dependent variables we re-run the same ANOVAs while entering the individual difference variables either as a further between subject factor or as a covariate. These analyses revealed that neither gender, age, performance anxiety, nor any of the sport-specific variables significantly influenced the main pattern of results.

Discussion

The central aim of the present research was to build and extend previous research^{15,20–24,32} in psychology by investigating the impact of evaluative feedback from coaches in the form of reward and dominant smiles in stressful tasks (speech and physical tasks) amongst athletes. To this end, we created a scenario in which athletes had to convince a coach in a speech about their physical and mental strengths and perform two complex physical tasks while being evaluated by a coach. Results showed that athletes who perceived dominant smiles as compared to reward smiles after their performances had a higher heart-rate than athletes perceiving rewarding feedback. In combination with previous research¹⁵ this is likely to occur via stress-related augmentation or dampening of the HPA axis activity as a consequence of either perceiving dominant smiles (i.e. increased HPA axis activity) or reward smiles (i.e. decreased HPA axis activity). In addition, the athletes receiving dominant feedback reported feeling less positive affect after their performances than those receiving rewarding feedback. No effects of the different feedback types were evident on subjective stress feelings and facial expressions of emotion as measured by Noldus FaceReader. In general, results are in line with the inferential processing mechanism described in the EASI model³² and with previous research suggesting that functionally distinct forms of human smiles have differential effects in social interactions.^{10,15}

The findings support psychological theorising and demonstrate the potential impact of non-verbal behaviour on humans.^{12–15,32} The present data might also be interpreted as in line with the behavioural ecology theory of facial expressions^{9,40} and the broader ecological theory of person perception^{41,42} that assumes that the perceptual abilities of humans have been shaped by their ecological utility. As a result, people have become especially attuned to facial expressions of other people in order to facilitate adaptive (social) behaviour. The present results indicated that the perception of different types of smiles in a coach affected

Table 2. Means and SDs for the Noldus FaceReader data as a function of measurement time and group and results for a 2 (baseline vs. average of feedback) $\times$ 2 (dominant vs. reward) ANOVA.

	Descriptive				Inferentiell		
	Before feedback		After feedback		<i>Interaction mt*group</i>		
	Reward <i>M (SD)</i>	Dominant <i>M (SD)</i>	Reward <i>M (SD)</i>	Dominant <i>M (SD)</i>			
					<i>F</i>	<i>P</i>	η^2
Action unit (intensity)							
Inner brow raiser	0.008 (0.02)	0.031 (0.07)	0.106 (0.00)	0.109 (0.09)	1.11	.30	.020
Outer brow raiser	0.004 (0.01)	0.015 (0.03)	0.066 (0.00)	0.065 (0.05)	1.35	.25	.024
Brow lowerer	0.026 (0.06)	0.012 (0.02)	0.012 (0.00)	0.013 (0.01)	1.61	.21	.028
Upper lid raiser	0.025 (0.05)	0.018 (0.03)	0.009 (0.00)	0.009 (0.00)	0.46	.50	.093
Cheek raiser	0.029 (0.04)	0.029 (0.04)	0.030 (0.00)	0.031 (0.02)	0.08	.78	.001
Lid tightener	0.019 (0.06)	0.011 (0.03)	0.009 (0.00)	0.008 (0.01)	0.32	.57	.006
Nose wrinkler	0.000 (0.00)	0.004 (0.02)	0.001 (0.00)	0.001 (0.00)	0.76	.39	.014
Upper lip raiser	0.010 (0.03)	0.004 (0.01)	0.010 (0.00)	0.010 (0.01)	1.37	.25	.024
Lip corner puller	0.079 (0.08)	0.085 (0.13)	0.046 (0.00)	0.047 (0.03)	0.02	.89	.000
Dimpler	0.004 (0.01)	0.003 (0.01)	0.003 (0.00)	0.003 (0.00)	0.13	.72	.002
Lip corner depressor	0.014 (0.05)	0.001 (0.00)	0.001 (0.00)	0.002 (0.00)	2.08	.16	.036
Chin raiser	0.059 (0.12)	0.009 (0.02)	0.003 (0.00)	0.003 (0.00)	5.14	.03*	.085
Lip puckerer	0.017 (0.06)	0.007 (0.02)	0.001 (0.00)	0.002 (0.00)	0.93	.34	.017
Lip stretcher	0.001 (0.00)	0.006 (0.02)	0.002 (0.00)	0.002 (0.00)	2.03	.16	.036
Lip tightener	0.006 (0.01)	0.004 (0.01)	0.002 (0.00)	0.002 (0.00)	0.12	.73	.002
Lip pressor	0.003 (0.01)	0.009 (0.03)	0.052 (0.00)	0.047 (0.03)	3.98	.05	.068
Lips part	0.082 (0.07)	0.114 (0.10)	0.064 (0.00)	0.067 (0.02)	1.41	.24	.025
Jaw drop	0.008 (0.02)	0.008 (0.02)	0.003 (0.00)	0.003 (0.00)	0.02	.89	.000
Mouth stretch	0.006 (0.03)	0.001 (0.00)	0.005 (0.00)	0.005 (0.01)	1.42	.24	.026
Eyes closed	0.156 (0.21)	0.202 (0.19)	0.047 (0.00)	0.046 (0.02)	0.77	.39	.014
Emotion							
Neutral	0.679 (0.12)	0.644 (0.10)	0.773 (0.00)	0.767 (0.05)	1.04	.31	.018
Happy	0.115 (0.07)	0.132 (0.08)	0.095 (0.00)	0.100 (0.03)	0.30	.58	.005
Sad	0.040 (0.03)	0.040 (0.04)	0.024 (0.00)	0.025 (0.01)	0.00	.96	.000
Angry	0.021 (0.03)	0.036 (0.04)	0.015 (0.00)	0.015 (0.01)	2.44	.12	.042
Surprised	0.012 (0.01)	0.016 (0.01)	0.017 (0.00)	0.017 (0.01)	2.54	.12	.044
Scared	0.015 (0.01)	0.021 (0.02)	0.011 (0.00)	0.011 (0.00)	2.71	.11	.047
Disgusted	0.013 (0.02)	0.011 (0.01)	0.011 (0.00)	0.010 (0.00)	0.17	.68	.003
Valence	0.044 (0.09)	0.047 (0.10)	0.055 (0.00)	0.060 (0.04)	0.01	.93	.000
Arousal	0.349 (0.10)	0.318 (0.07)	0.453 (0.00)	0.454 (0.03)	1.65	.21	.029

an athlete's heart rate and affect and thereby would likely impact their future behaviour. In this respect, it seems likely that the perception of reward smiles from a coach induced pleasurable states in the athletes and thereby reinforced the behaviour that gave rise to the smile initially. In the case of the present study and in line with the EASI theorising, athletes likely inferred that the coach was happy with their performance and thereby reinforced their behaviour. On the other hand, athletes observing dominant smiles subsequent to their performances likely inferred that they had not performed very well, which resulted in negative affective reactions. In this respect, it seems likely that dominant smiles shown by coaches can be used as a tool for managing hierarchical relationships in coach-athlete dyads or teams, for example, to subdue athletes in lower-status positions compared to the coach.⁴ In the case of the

present study, athletes likely inferred that the coach was scornful of their performance and showed his disrespect.

The present study extends initial research in the field of sports that has initiated the investigating of non-verbal behaviour in coaches.^{17–22} While this previous research mainly investigated how an athlete's behaviour and performance was affected by a coach's emotional behaviour, the present research was the first to show that even very subtle differences in a coach's face have the potential to affect an athlete's physiology and current affective state when trying to perform at their best. In this respect, the present research points to a potential mechanism of how a coach's emotional expressions might impact athletes, that is, by affecting HPA axis activity.¹⁵ As the HPA has been shown to control bodily reactions to stress, it seems feasible that a coach's facial expressions have the potential to calm

athletes down and help them cope with anxiety and stress that coincide with sports performance.

Moreover, the results indicate that a dominant smile by a coach has the potential to upset an athlete and potentially will affect the coach–athlete relationship. Given the importance that has been attributed to the coach–athlete relationship,⁴³ the present research can contribute to this literature as it indicates that even subtle facial expressions by a coach can have an impact on the athletes. Although this remains speculative at present, it seems likely that dominant smiles by a coach might have the potential to negatively impact the behaviour, motivation, and performance of athletes. On the other hand, a rewarding smile by a coach has the potential to have beneficial effects on an athlete and might positively affect their behaviour. In this respect, the present research extends prior research on the effects of verbal feedback^{23–27} and emotional expressions of coaches on athletes.^{20–24,28–31} Hence, future research might want to continue this line of investigation by investigating the effects of coaches' facial expressions on the coach–athlete relationship and, in turn, sports performance as it seems likely that facial expressions shown by coaches, in the short- and long-run, have the potential to affect various important outcomes in sports.

Limitations

While the present research extends the growing body of literature on non-verbal behaviour in sports (see Refs.^{44,45} for reviews), the present research has limitations. First, the present research was only sufficiently powered to detect medium-to-large effects and therefore it is possible that smaller effects were not detected. Moreover, we cannot be entirely sure that changes in heart-rate are indeed a result of HPA axis activity caused by a stress response. Although, this pattern of results fits previous research and theorising,¹⁵ we cannot entirely rule out that participants observing the dominant smile were motivated to put more effort in than the group observing the rewarding smiles. However, we consider this fairly unlikely as the differences in heart-rate were also accompanied by differences in perceived affect. In addition, it remains unclear if the effects found are specific to the particular experimental context or if these would transfer to non-athlete populations and different evaluative performance contexts.

Further, the present design might be limited in its external validity that resulted from the high experimental control in the design of the study. As the coach was not the actual coach of the athletes and the feedback was simulated via pre-recorded video material in a Skype video stream, it is possible that some athletes did not pay much attention to the facial expressions or did not care much about them. Therefore, the artificial laboratory situation is certainly a limitation of the study. However, in the debriefing at the end of the experiment, none of the athletes reported that they had noticed that they were viewing recordings of the

coach and believed that they were viewing a live stream. In addition, future research might benefit from including a neutral control condition in which a coach shows a neutral facial expression as opposed to emotionally valenced expressions to scrutinise the assumed facilitative effects of a reward smile and the debilitative effects of a dominant smile. Nevertheless, we consider it plausible that the effects of non-verbal facial feedback would, in fact, be enhanced in more representative settings of coach–athlete interactions as emotional expressions and physical proximity would likely be more intense in these situations. However, we acknowledge that these first findings need to be replicated in representative sports settings, while further improving methodological procedure and reducing experimenter errors.

Conclusion

In conclusion, the present results are the first to show that subtle differences in facial expressions of coaches can affect an athlete's heart rate and affective experience. Arguably, this finding indicates that coaches need to pay attention to their non-verbal behaviour when coaching athletes, as this might impact the coach–athlete relationship and the performance of the athlete.

Data availability

All data will be made available at reasonable request to the corresponding author.

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